# 1. Introduction

In retail sectors, understanding the factors that influence sales is crucial to make strategic decisions. This project analyzes the Carseats dataset, which contains information from 400 stores, including pricing, demographics, shelving quality, advertising, and other store attributes. The goal is to identify key determinants of unit sales and quantify their effects. This analysis is practically meaningful for retail management as it helps to answer these questions:

- Which store characteristics lead to higher performance?
- Do demographic features meaningfully impact sales?
- Is there a difference in the distribution of advertising level across regions?
- Are consumers sensitive to price?

All analyses are conducted in Python, using libraries such as:

- Pandas for data manipulation
- Numpy for numerical operations
- Matplotlib/ Seaborn for visualisation
- Statsmodel for regression modeling
- Scipy for statistical test

Below are the libraries that are used in order to perform EDA (Exploratory data analysis)

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf
from scipy import stats
```

# 2. Dataset overview

In [2]:
```
df=pd.read_excel("Carseats.xlsx")
```

### 2.1. General description of the data

We begin the analysis by examining the basic structure of the dataset, starting with its shape. This helps us understand how many observations and variables are available for analysis.

In [3]:
```
# Shape
print("\n1.DATASET SHAPE:", df.shape)
```

```
1.DATASET SHAPE: (400, 11)
```

In [4]:
```
# Basic info
print("\n2. DATASET INFO:")
df.info()
```

```
2. DATASET INFO:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 11 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Sales        399 non-null    float64
 1   CompPrice    399 non-null    float64
 2   Income       399 non-null    float64
 3   Advertising  400 non-null    int64
 4   Population   400 non-null    int64
 5   Price        400 non-null    int64
 6   ShelveLoc    400 non-null    object
 7   Age          400 non-null    int64
 8   Education    400 non-null    int64
 9   Urban        399 non-null    object
 10  US           400 non-null    object
dtypes: float64(3), int64(5), object(3)
memory usage: 34.5+ KB
```

The dataset contains 400 observations and 11 variables including both numerical and categorical variables. Among 11 variable, we have 6 numerical variable, 3 categorical variables and 2 integer demographic variables. Variable description:

1. Sales: Unit sales (in thousands) at each location
2. CompPrice: Price charged by competitor at each location
3. Income: Community income level (in thousands of dollars)
4. Advertising: Local advertising budget for company at each location (in thousands of dollars)
5. Population: Population size in region (in thousands)

6. Price: Price company charges for car seats at each site
7. ShelveLoc: A factor with levels Bad, Good and Medium indicating the quality of the shelving location for the car seats at each site
8. Age: Average age of the local population
9. Education: Education level at each location
10. Urban: A factor with levels No and Yes to indicate whether the store is in an urban or rural location
11. US: A factor with levels No and Yes to indicate whether the store is in the US or not

Most columns have complete data, except:

- Sales have 1 missing value
- CompPrice have 1 missing value
- Income have 1 missing value
- Urban have 1 missing value

Below are the first five rows of the dataset, providing an initial overview of all variables.

In [5]:
```
print("\n3. First 5 rows:")
df.head()
```

```
3. First 5 rows:
```

Out[5]:

| | Sales | CompPrice | Income | Advertising | Population | Price | ShelveLoc | Age | Education | Urban | US |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 9.50 | 138.0 | 73.0 | 11 | 276 | 120 | Bad | 42 | 17 | Yes | Yes |
| **1** | 11.22 | 111.0 | 48.0 | 16 | 260 | 83 | Good | 65 | 10 | Yes | Yes |
| **2** | 10.06 | 113.0 | 35.0 | 10 | 269 | 80 | Medium | 59 | 12 | Yes | Yes |
| **3** | 7.40 | 117.0 | 100.0 | 4 | 466 | 97 | Medium | 55 | 14 | Yes | Yes |
| **4** | 4.15 | 141.0 | 64.0 | 3 | 340 | 128 | Bad | 38 | 13 | Yes | No |

### 2.2. Descriptive statistics

In [6]:
```
# Descriptive statistics
print("\n4. BASIC STATISTICS:")
df.describe()
```

```
4. BASIC STATISTICS:
```

Out[6]:

| | Sales | CompPrice | Income | Advertising | Population | Price | Age | Education |
|---|---|---|---|---|---|---|---|---|
| **count** | 399.000000 | 399.000000 | 399.000000 | 400.000000 | 400.000000 | 400.000000 | 400.000000 | 400.000000 |
| **mean** | 7.504035 | 124.967419 | 68.639098 | 6.635000 | 264.840000 | 115.795000 | 53.322500 | 13.900000 |
| **std** | 2.823442 | 15.353013 | 28.018750 | 6.650364 | 147.376436 | 23.676664 | 16.200297 | 2.620528 |
| **min** | 0.000000 | 77.000000 | 21.000000 | 0.000000 | 10.000000 | 24.000000 | 25.000000 | 10.000000 |
| **25%** | 5.410000 | 115.000000 | 42.500000 | 0.000000 | 139.000000 | 100.000000 | 39.750000 | 12.000000 |
| **50%** | 7.490000 | 125.000000 | 69.000000 | 5.000000 | 272.000000 | 117.000000 | 54.500000 | 14.000000 |
| **75%** | 9.320000 | 135.000000 | 91.000000 | 12.000000 | 398.500000 | 131.000000 | 66.000000 | 16.000000 |
| **max** | 16.270000 | 175.000000 | 120.000000 | 29.000000 | 509.000000 | 191.000000 | 80.000000 | 18.000000 |

The dataset includes 400 observations covering socioeconomic and marketing variables. Sales vary widely across regions (mean 7,504 units; range 0–16,270 units). Competitive price is relatively stable (mean 125 USD and range from 77,000 USD to 175,000 USD), while product price shows much larger variation (24 – 191 USD). Income also varies significantly (mean 68,640 USD and standard deviation of 27,983 USD), indicating diverse economic conditions. Advertising spending is low on average (6,635 USD) but highly inconsistent (standard deviation 6,650 USD). Population ranges widely, reflecting demographic diversity. The average age is 53.32 years, and education levels range from 10 to 18 years (mean 13.9).

## 3. Data quality check

### 3.1. Missing values

Before proceeding with further analysis, it is important to check the dataset for missing values.

In [7]:
```
# Check missing values
print("\n5. MISSING VALUES:")
df.isna().sum()
```

```
5. MISSING VALUES:
```

Out[7]:
```
Sales          1
CompPrice      1
Income         1
Advertising    0
Population     0
Price          0
ShelveLoc      0
Age            0
Education      0
Urban          1
US             0
dtype: int64
```

Given that only a small number of entries are missing (one value in Sales, CompPrice, Income, and Urban), these observations can be safely handled without significantly affecting the analysis.

In [8]:
```
# Handling missing values:
# Continuous variable
df['Sales']     = df['Sales'].fillna(df['Sales'].median())
df['CompPrice'] = df['CompPrice'].fillna(df['CompPrice'].median())
df['Income']    = df['Income'].fillna(df['Income'].median())
# Categorical variable
df['Urban']     = df['Urban'].fillna(df['Urban'].mode()[0])
```

To handle missing values, some strategies are applied depending on type of variable:

- Handling to continuous variable: for numerical variable, the median value of each columnn is used to filling because median is more robust to outliners and maintains the central tendency of the distribution.
- Handling to categorical variable: for the categorical variable Urban, the missing value is filled by the mode as the most most frequent category in that column.

In [9]:
```
# Check missing values after handling
df.isnull().sum()
```

Out[9]:
```
Sales          0
CompPrice      0
Income         0
Advertising    0
Population     0
Price          0
ShelveLoc      0
Age            0
Education      0
Urban          0
US             0
dtype: int64
```

**3.2. Duplicated values**

Checking for duplicate records is an essential step in Exploratory Data Analysis because duplicated observations can distort the statistical structure of the dataset and lead to incorrect analytical conclusions.

In [10]:
```
# Check duplicated values:
print("\n6. Duplicated values:")
df.duplicated().sum()
```

```
6. Duplicated values:
```

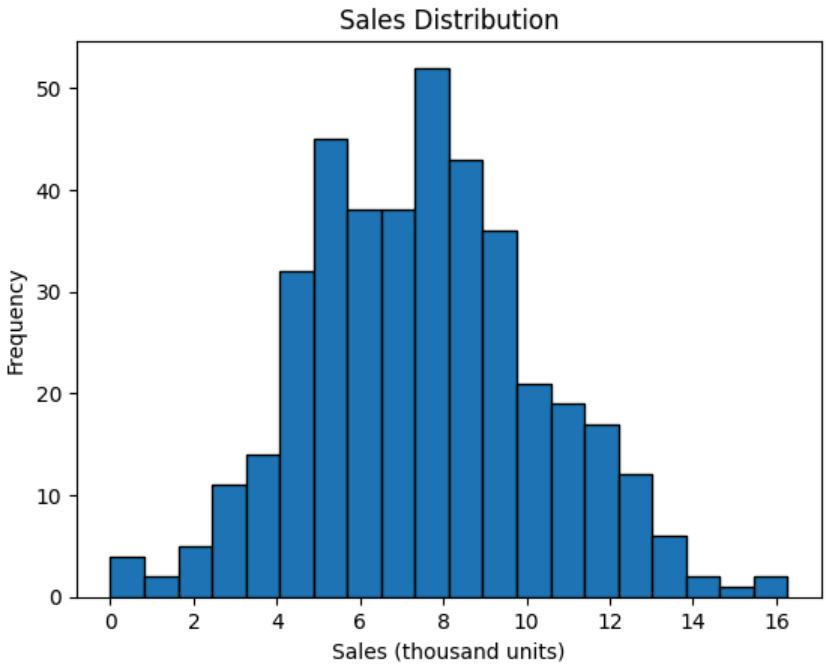
Out[10]:
```
np.int64(0)
```

There is no duplicate data in this dataset

# 4. Visualisation and its interpret

**A. The distribution of one or more individual continuous variables**

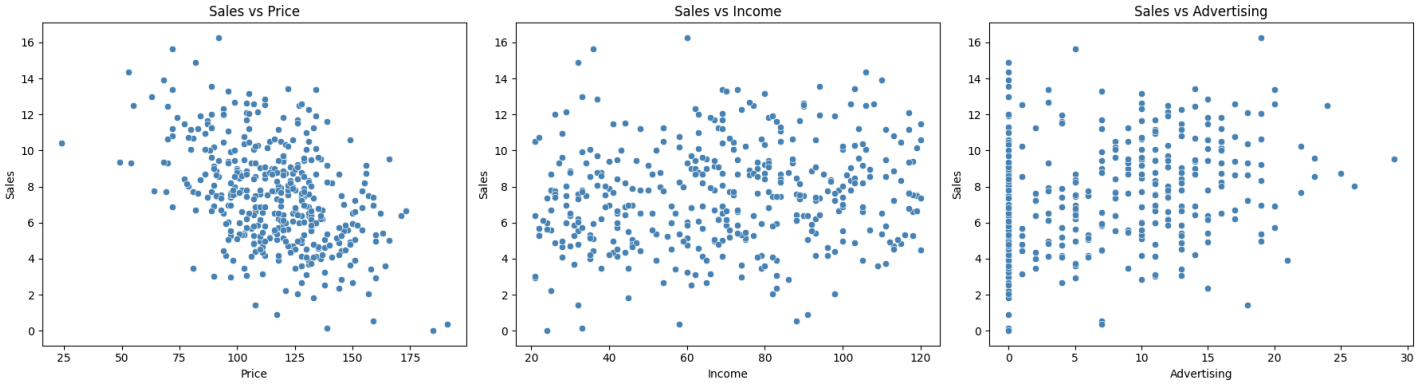
In [11]:
```
# Sales distribution
df['Sales'].hist(bins=20, edgecolor='black')
plt.title('Sales Distribution')
plt.xlabel('Sales (thousand units)')
plt.ylabel('Frequency')
plt.grid ( False)
```




Histogram is used to describe and visualise the distribution of the Sales variable.The histrogram shows that the sales values range from 0 to 16,000 units, forming a roughly bell-shaped distribution with a slight positive (right) skewness. Most observations cluster around the 6,000-10,000 units range which represents the highest frequency region in the distribution. In a positively skewed distribution, these higher-end observations tend to pull the mean upward, making it slightly higher than both the median and the most commonly occurring sales range.

**B. The relationship of a pair of continuous variables**

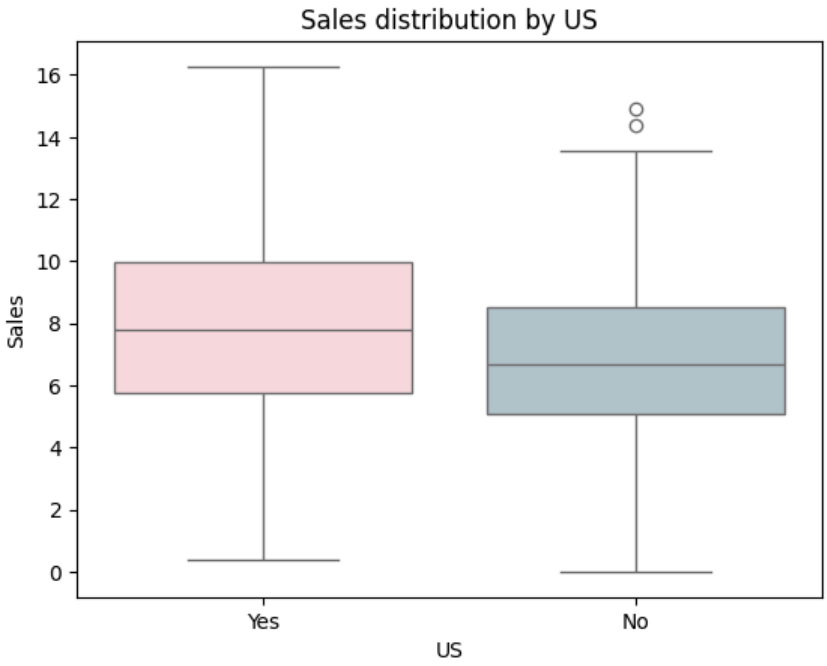
In [12]:
```
num_vars = ["Price", "Income", "Advertising"]
fig, axes = plt.subplots(1, 3, figsize=(18, 5))
axes = axes.flatten()
for i, var in enumerate(num_vars):
    sns.scatterplot(data=df, x=var, y="Sales", ax=axes[i], color="steelblue")
    axes[i].set_title(f"Sales vs {var}")
    axes[i].set_xlabel(var)
    axes[i].set_ylabel("Sales")
plt.tight_layout()
plt.show()
```




The scatter plots illustrate the relationship Sales with Price, Income, Advertising. The relationship between price and sales exhibit a clear negative pattern, where higher prices are generally linked to lower sales despite some dispersion. Income and Sales show wide variability, with only a slight tendency for sales to increase in higher-income areas. The relationship between Advertising and Sales display a mild upward trend, indicating that greater advertising expenditure is associated with slightly higher sales.

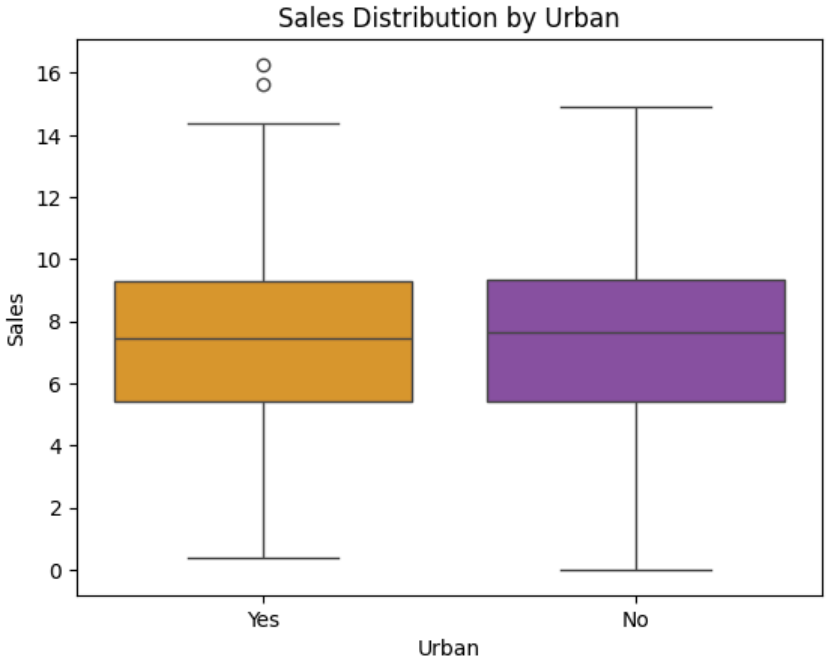
**C. The association between a categorical variable and a continuous one**

In [13]:
```
# Boxplot between US and Sales
sns.boxplot(data=df, x='US', y='Sales', hue='US', palette=['#FFD1DC', '#AEC6CF'])
plt.title("Sales distribution by US")
plt.xlabel("US")
plt.ylabel("Sales")
plt.show()
```



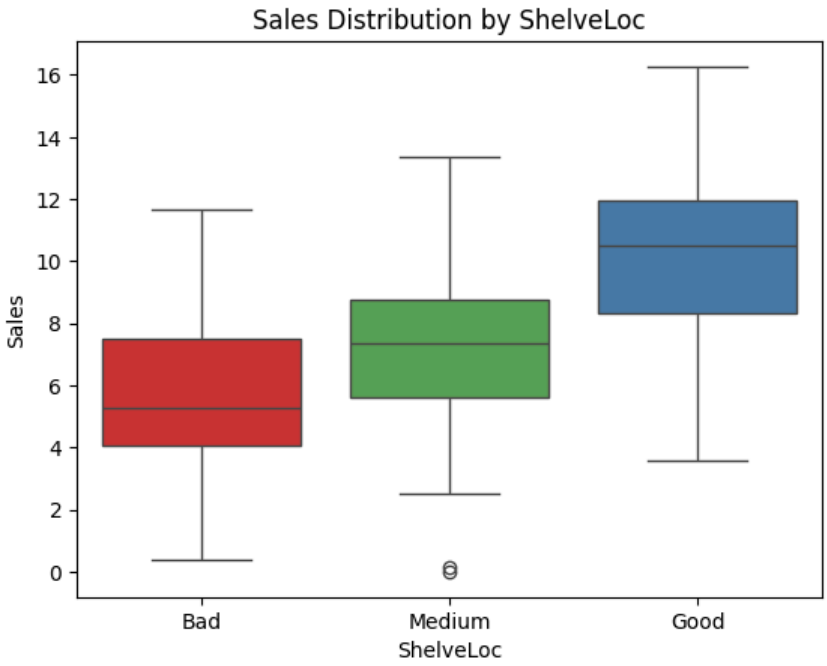

The boxplot comparing the distribution of Sales between the US and non-US groups reveals a clear distinction in Sales across regions. The US group has a significantly higher median value than the non-US group and a wider distribution that includes several high-performing observations. This reflects that the US market not only has higher average sales but also exhibits greater diversity in business results. In contrast, the non-US group has generally lower sales levels although some high outliners are present.

In [14]:
```
# Boxplot between Urban and Sales
sns.boxplot(data=df, x='Urban', y='Sales',hue='Urban',palette=['#f39c12', '#8e44ad'])
plt.title("Sales Distribution by Urban")
plt.xlabel("Urban")
plt.ylabel("Sales")
plt.show()
```




These boxplots indicate that the distribution of sales for urban and non-urban settings is almost identical, with nearly equivalent medians, IQRs, and overall ranges. While there are a few high outliers for the urban group, both settings have similar sales patterns, demonstrating that an urban location does not strongly impact sales performance.

In [15]:
```
# Boxplot between ShelveLoc and Sales
sns.boxplot(data=df, x="ShelveLoc", y="Sales", hue = "ShelveLoc", order=["Bad", "Medium", "Good"],palette=['#e41a1c', '#377eb8', '
plt.title("Sales Distribution by ShelveLoc")
plt.show()
```




The boxplot clearly illustrates that there are differences in sales among the three ShelveLoc categories. Good shelf locations have the highest sales, with the strongest median and widest spread. Medium shelf locations have a moderate median, higher than Bad but lower than Good. Poor shelf placement, on the other hand, has the lowest median sales with a distribution concentrated at the lower end, proving that poor product placement significantly reduces sales performance.

**D. The relationship between more than two variables**

**The relationship between Price and Sales by ShelveLoc**

In [16]:
```
sns.scatterplot(data=df, x="Price", y="Sales", hue="ShelveLoc",palette="Set1")
plt.title("The relationship between Price and Sales by ShelveLoc")
plt.xlabel("Price (USD)")
plt.ylabel("Sales (thousand units)")
plt.legend(title="ShelveLoc")
plt.grid(False)
plt.show()
```




The scatter plot shows a clear negative relationship between Sales and Price where sales tend to increase as price decreases. However, the sales level is also strongly affected by the display location. Products located in the Good position always achieve significantly higher sales than "Medium" and " Bad" even at similar price levels. This suggests that shelf placement plays a crucial role in driving sales and can have a greater impact than price adjustment alone.

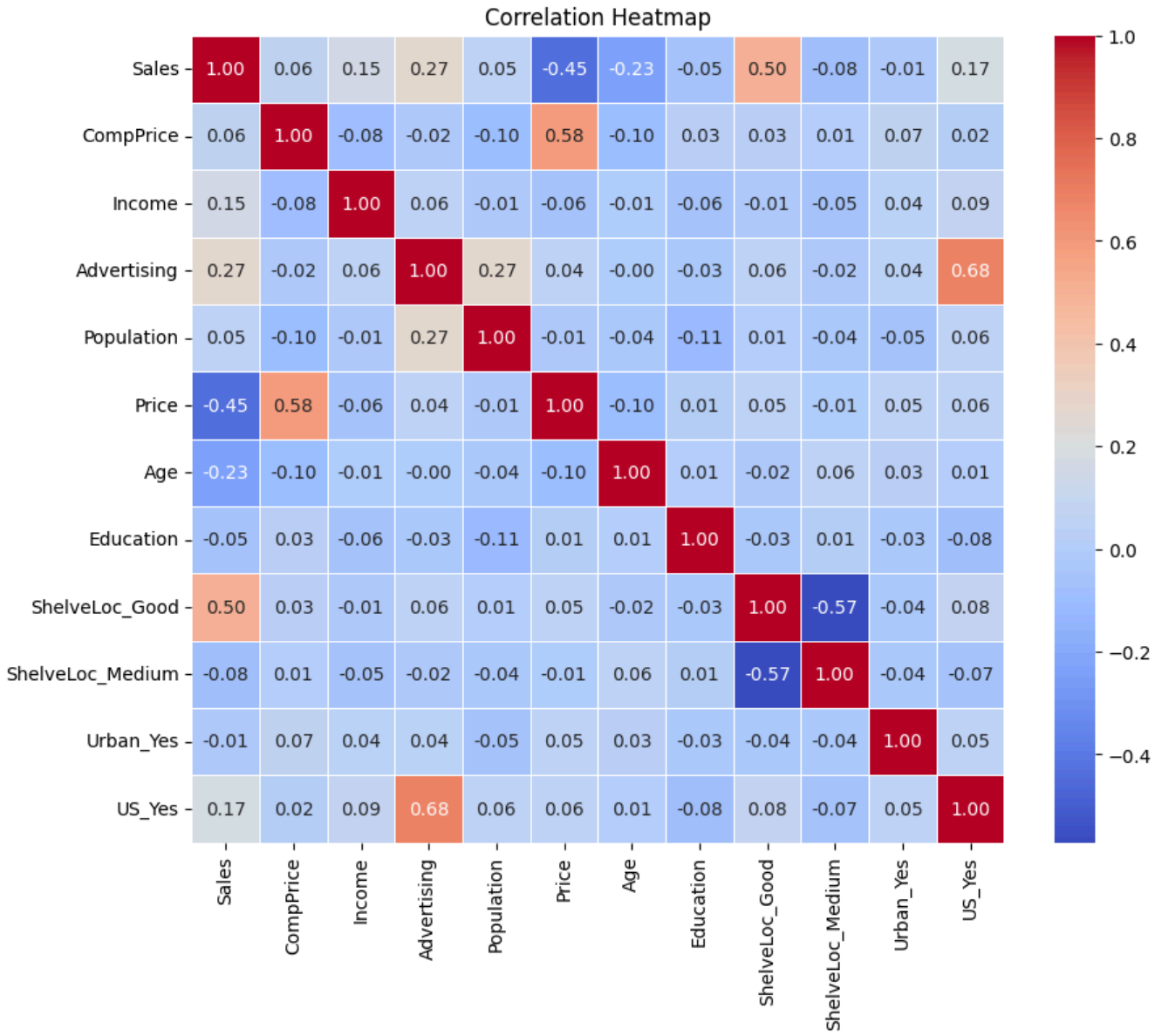
**Correlation heatmap**

There are three categorical variables (ShelveLoc, Urban, US). Since the correlation matrix can only calculated between numerical variables, before drawing heatmap, the categorical variables are converted.

In [17]:
```
cat_cols = ["ShelveLoc", "Urban", "US"]

df_encoded = pd.get_dummies(df, columns=cat_cols, drop_first=True)
```

In [18]:
```
# Correlation heatmap
plt.figure(figsize=(10, 8))
ols_vars = ['Sales', 'CompPrice', 'Income', 'Advertising','Population', 'Price', 'Age', 'Education','ShelveLoc_Good',
            'ShelveLoc_Medium','Urban_Yes', 'US_Yes']
corr = df_encoded[ols_vars].corr()
sns.heatmap(corr, annot=True, fmt=".2f",cmap="coolwarm",linewidths=0.5)
plt.title("Correlation Heatmap")
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.show()
```




The correlation heatmap shows that Sales is most strongly associated with ShelveLoc_Good and Price. Specifically, Good shelf placement increases Sales while higher prices reduce Sales. Advertising and Income show weak positive relationship with Sales and other variables have minimal linear impact.

## 5. Display unique values of categorical variable and their frequencies

In [19]:
```
for col in df.select_dtypes(include=['object']).columns:
    print(f"--- {col} ---")
    print(df[col].value_counts())
    print("\n")
```

```
--- ShelveLoc ---
ShelveLoc
Medium    219
Bad        96
Good       85
Name: count, dtype: int64


--- Urban ---
Urban
Yes    282
No     118
Name: count, dtype: int64


--- US ---
US
Yes    258
No     142
Name: count, dtype: int64
```

The data shows noticeable imbalance across the categorical variables. For the ShelveLoc variable, The Medium shelf location appears the most with 219 cases while Bad and Good are 96 and 85 cases respectively, showing that most of the products are placed in the Medium position and Good shelf location is quite limited. The Urban variable is strongly skewed toward urban areas with 282 observations compared to 118 from non-urban areas. Similarly, the US variable contains more customers from the US than from outside the US.

## 6. Build a contingency table of two potentially related categorical variables

To examine whether advertising behaviour differs between US and non-US stores, the Advertising variable is split into High and Low based on the median in order to create a balanced segmentation of stores based on their advertising intensity.Because advertising expenditure can vary widely across stores and may include several extreme values, the median provides a robust and non-biased threshold that is less affected by outliers than the mean. This segmentation allows for a clearer comparision across regions.

In [20]:
```
# Creating advertising level
median_adv = df["Advertising"].median()
df["AdvLevel"] = df["Advertising"].apply(lambda x: "Low" if x < median_adv else "High")
print (median_adv)
```

```
5.0
```

Let's look at the contingency table of two potentially related categorical variables

In [21]:
```
# Contingency table
cont_table = pd.crosstab(df["AdvLevel"], df["US"])
print(cont_table)
```

```
US         No  Yes
AdvLevel
High        6  207
Low       136   51
```

**Step 1: Formulate the null hypothesis and alternative hypotheis**

(Null hypothesis) H0: There is no association relationship between Advertising level and US

(Alternative hypothesis) H1: There is an association relationship between Advertising level and US

**Step 2: Test statistic: Chi-square**

**Step 3: Choose level of significance: 95%**

**Step 4: Calculate the test statistic**

Let's determine the value of two categorical variables so the test to use is the Chi-square test of independence

In [22]:
```
# chi-square test
chi2, p_val,dof, expected = stats.chi2_contingency(cont_table)
print(f"p-value:{p_val}")
```

```
p-value:1.7610098484045262e-47
```

Technical conclusion p-value < 0.05 hence we reject the null hypothesis. Therefore, there is strong evidence to believe that there is statistically siginificant evidence at the 5% level to conclude that advertising levels and US are associated.

## 7. Retrieve one or more subsets of rows based on two or more logical criteria

Income and Price are converted into “High” and “Low” categories to simplify comparison and uncover behavioral patterns by using their median values in order to create balanced customer segments.The median is a robust statistical threshold that divides the dataset into two groups of approximately equal size, ensuring that neither category is overly dominant.Combining them creates four segments: High income–High price, High

income–Low price, Low income–High price, and Low income–Low price. These groups represent different customer profiles and allow a clearer analysis of how income levels relate to price sensitivity and purchasing behavior.

In [23]:
```
# Create Income groups
income_threshold = df["Income"].median()
df["IncomeGroup"] = df["Income"].apply(lambda x: "High Income" if x > income_threshold else "Low Income")
print(income_threshold)
```

```
69.0
```

In [24]:
```
# Create Price groups
price_threshold = df["Price"].median()
df["PriceGroup"] = df["Price"].apply(lambda x: "High Price" if x > price_threshold else "Low Price")
```

**High income group**

In [25]:
```
# Create subsets for high income
high_income_high_price = df[(df["IncomeGroup"]=="High Income") & (df["PriceGroup"]=="High Price")]
high_income_low_price = df[(df["IncomeGroup"]=="High Income") & (df["PriceGroup"]=="Low Price")]
```

In [26]:
```
print("=== High Income - High Price ===")
display(high_income_high_price.describe().T)

print("\n=== High Income - Low Price ===")
display(high_income_low_price.describe().T)
```

```
=== High Income - High Price ===
```

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **Sales** | 90.0 | 7.073444 | 2.661398 | 0.53 | 5.1025 | 6.86 | 8.7475 | 13.44 |
| **CompPrice** | 90.0 | 130.777778 | 13.160870 | 96.00 | 122.0000 | 131.50 | 138.0000 | 161.00 |
| **Income** | 90.0 | 93.977778 | 14.693098 | 70.00 | 81.0000 | 93.00 | 105.0000 | 120.00 |
| **Advertising** | 90.0 | 6.922222 | 6.957702 | 0.00 | 0.0000 | 5.00 | 12.0000 | 25.00 |
| **Population** | 90.0 | 264.911111 | 145.915140 | 10.00 | 137.5000 | 267.00 | 406.0000 | 500.00 |
| **Price** | 90.0 | 133.911111 | 12.847983 | 118.00 | 124.0000 | 130.00 | 144.7500 | 164.00 |
| **Age** | 90.0 | 49.577778 | 16.118227 | 25.00 | 35.0000 | 48.00 | 63.0000 | 80.00 |
| **Education** | 90.0 | 13.744444 | 2.786650 | 10.00 | 11.0000 | 13.00 | 16.7500 | 18.00 |

```
=== High Income - Low Price ===
```

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **Sales** | 103.0 | 8.567379 | 2.733625 | 0.91 | 6.795 | 8.47 | 10.485 | 14.37 |
| **CompPrice** | 103.0 | 116.514563 | 14.023694 | 85.00 | 106.500 | 118.00 | 125.000 | 148.00 |
| **Income** | 103.0 | 92.330097 | 15.012014 | 70.00 | 79.000 | 90.00 | 104.500 | 120.00 |
| **Advertising** | 103.0 | 6.553398 | 6.321033 | 0.00 | 0.000 | 6.00 | 12.000 | 23.00 |
| **Population** | 103.0 | 265.582524 | 156.117259 | 16.00 | 130.000 | 266.00 | 404.000 | 509.00 |
| **Price** | 103.0 | 96.359223 | 15.597168 | 49.00 | 89.000 | 99.00 | 108.000 | 117.00 |
| **Age** | 103.0 | 57.135922 | 15.325263 | 25.00 | 47.000 | 60.00 | 68.000 | 80.00 |
| **Education** | 103.0 | 13.766990 | 2.672312 | 10.00 | 11.000 | 14.00 | 16.000 | 18.00 |

Although the two groups share similar income levels, their purchasing behavior differs notably. The High Income–High Price group records lower average sales (about 7,073 units), while the High Income–Low Price group achieves higher sales (around 8,567 units), indicating that even high-income customers buy more when prices are lower. Despite having a larger population and lower advertising spending, the low-price group still performs better, suggesting that price plays a more important role than advertising in driving sales within high-income areas.

**Step 1: Formulate the null hypothesis and alternative hypothesis**

Null hypothesis (H0): There is no difference in sales between High price and Low price within the High income group

Alternative hypothesis (H1): There is a difference in sales such that the mean sales in High price are significantly lower than the mean sales in Low price within the High income group.

**Step 2: Test statistic: Independent sample t-test**

**Step 3: Choose level of significance: 95%**

**Step 4: Calculate the test statistic**

In [27]:
```
t_val, p_two = stats.ttest_ind(high_income_high_price["Sales"], high_income_low_price["Sales"])
p_one = p_two/2 if t_val < 0 else 1 - p_two/2
print (f"t-value: {t_val},one-tailed p-value:{p_one}")
```

```
t-value: -3.834380422959029,one-tailed p-value:8.542488157327464e-05
```

Technical conclusion: p-value < 0.05 so we reject the null hypothesis. There is a difference in sales such that the mean sales in High price are significantly lower than the mean sales in Low price within the High income group.

**Low income group**

In [28]:
```
# Create subsets for low income
low_income_high_price = df[(df["IncomeGroup"]=="Low Income") & (df["PriceGroup"]=="High Price")]
low_income_low_price = df[(df["IncomeGroup"]=="Low Income") & (df["PriceGroup"]=="Low Price")]
```

In [29]:
```
print("=== Low Income - High Price ===")
display(low_income_high_price.describe().T)

print("\n=== Low Income - Low Price ===")
display(low_income_low_price.describe().T)
```

```
=== Low Income - High Price ===
```

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **Sales** | 109.0 | 6.356055 | 2.517226 | 0.0 | 4.74 | 6.15 | 7.96 | 13.39 |
| **CompPrice** | 109.0 | 133.477064 | 13.301766 | 102.0 | 124.00 | 132.00 | 141.00 | 175.00 |
| **Income** | 109.0 | 45.972477 | 15.233649 | 21.0 | 33.00 | 44.00 | 61.00 | 69.00 |
| **Advertising** | 109.0 | 6.899083 | 6.810864 | 0.0 | 0.00 | 6.00 | 12.00 | 29.00 |
| **Population** | 109.0 | 262.091743 | 136.954138 | 16.0 | 148.00 | 276.00 | 366.00 | 507.00 |
| **Price** | 109.0 | 135.174312 | 14.678558 | 118.0 | 125.00 | 132.00 | 139.00 | 191.00 |
| **Age** | 109.0 | 51.761468 | 15.840583 | 25.0 | 40.00 | 53.00 | 64.00 | 80.00 |
| **Education** | 109.0 | 14.211009 | 2.442196 | 10.0 | 12.00 | 14.00 | 16.00 | 18.00 |

```
=== Low Income - Low Price ===
```

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **Sales** | 98.0 | 8.058571 | 2.851868 | 1.42 | 5.895 | 7.8 | 9.8725 | 16.27 |
| **CompPrice** | 98.0 | 119.051020 | 13.435123 | 77.00 | 111.000 | 118.5 | 128.0000 | 146.00 |
| **Income** | 98.0 | 45.683673 | 15.042618 | 21.00 | 32.250 | 44.0 | 60.0000 | 69.00 |
| **Advertising** | 98.0 | 6.163265 | 6.592238 | 0.00 | 0.000 | 4.5 | 11.0000 | 20.00 |
| **Population** | 98.0 | 267.051020 | 152.580104 | 12.00 | 133.750 | 284.0 | 403.0000 | 508.00 |
| **Price** | 98.0 | 98.030612 | 14.569743 | 24.00 | 90.000 | 101.0 | 108.7500 | 117.00 |
| **Age** | 98.0 | 54.489796 | 16.800985 | 25.00 | 41.500 | 55.5 | 68.7500 | 80.00 |
| **Education** | 98.0 | 13.836735 | 2.611212 | 10.00 | 11.000 | 14.0 | 16.0000 | 18.00 |

The descriptive statistics for the Low Income – High Price and Low Income – Low Price groups reveal a clear influence of pricing on purchasing behavior among low-income customers. The Low Income – High Price group shows a lower average sales level (about 6356 units) compared to the Low Income – Low Price group (8059 units). This difference indicates that low-income consumers are highly sensitive to price changes, with lower prices driving noticeably higher sales.

**Step 1: Formulate the null hypothesis and alternative hypothesis**

Null hypothesis (H0): There is no difference in sales between High price and Low price within the Low income group

Alternative hypothesis (H1): There is a difference in sales such that the mean sales in High price are significantly lower than the mean sales in Low price within the Low income group.

**Step 2: Test statistic: Independent sample t-test**

**Step 3: Choose level of significance: 95%**

**Step 4: Calculate the test statistic**

In [30]:
```
t_val,p_val = stats.ttest_ind(low_income_high_price["Sales"], low_income_low_price["Sales"])
p_one = p_val/2 if t_val < 0 else 1 - p_val/2
print(f"t-value:{t_val},p-value:{p_one}")
```

```
t-value:-4.56216668038318,p-value:4.35490239784933e-06
```

Technical conclusion: p-value < 0.05 so we reject the null hypothesis. So there is strong evidence to conclude that there is a difference in sales such that the mean sales in High price are significantly lower than the mean sales in Low price within the Low income group.

# 8. Grouped analysis

To evaluate how Sales and pricing vary across shelf placement categories, we group the data by ShelveLoc and compute summary statistics for Sales and Price.

In [31]:
```
grouped_stats = df.groupby("ShelveLoc").agg({"Sales": ["mean", "sum"],"Price": "mean"})
grouped_stats
```

Out[31]:

| | Sales | | Price |
|---|---|---|---|
| | mean | sum | mean |
| **ShelveLoc** | | | |
| **Bad** | 5.554896 | 533.27 | 114.270833 |
| **Good** | 10.214000 | 868.19 | 117.882353 |
| **Medium** | 7.306575 | 1600.14 | 115.652968 |

The statistics table shows that the position of the product display has a great impact on sales performance. Stores with Good shelf locations achieve the highest average sales (about 10,214 units), significantly outperforming stores with Medium locations ( approximately 7,307 units) and Bad locations ( about 5,555 units). Although the Good group has the highest average price (117,88 USD/seat), their sales still far exceed the other groups, showing that the quality of the display locations is more important than the impact of the price. The Bad group has the lowest average price (114,27 USD/seat) but sales are still the lowest, reinforcing the conclusion that displaying products in a favourable location plays a major role in promoting buying behavior.

## 9. Linear regression analysis

Before fitting the linear regression model, the categorical variables in the dataset need to be transformed into a numerical format.

In [32]:
```
df_encoded = pd.get_dummies( df, columns=["ShelveLoc", "Urban", "US"],drop_first=True)
df_encoded.head()
```

Out[32]:

| | Sales | CompPrice | Income | Advertising | Population | Price | Age | Education | AdvLevel | IncomeGroup | PriceGroup | ShelveLoc_Good | ShelveLo |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 9.50 | 138.0 | 73.0 | 11 | 276 | 120 | 42 | 17 | High | High Income | High Price | False | |
| **1** | 11.22 | 111.0 | 48.0 | 16 | 260 | 83 | 65 | 10 | High | Low Income | Low Price | True | |
| **2** | 10.06 | 113.0 | 35.0 | 10 | 269 | 80 | 59 | 12 | High | Low Income | Low Price | False | |
| **3** | 7.40 | 117.0 | 100.0 | 4 | 466 | 97 | 55 | 14 | Low | High Income | Low Price | False | |
| **4** | 4.15 | 141.0 | 64.0 | 3 | 340 | 128 | 38 | 13 | Low | Low Income | High Price | False | |

The goal of the OLS model is to analyze the influence of factors such as competitive price, income group, advertising level, price group, age and shelf position on sales.

In [33]:
```
formula = "Sales ~ CompPrice + Income + Advertising + Price + Age + ShelveLoc"
model = smf.ols(formula=formula, data=df).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.869
Model:                            OLS   Adj. R-squared:                  0.866
Method:                 Least Squares   F-statistic:                     369.9
Date:                Wed, 04 Nov 2026   Prob (F-statistic):          2.28e-168
Time:                        21:45:56   Log-Likelihood:                -575.97
No. Observations:                 400   AIC:                             1168.
Df Residuals:                     392   BIC:                             1200.
Df Model:                           7
Covariance Type:            nonrobust
==============================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                  5.4684      0.511     10.701      0.000       4.464       6.473
ShelveLoc[T.Good]          4.8051      0.154     31.139      0.000       4.502       5.109
ShelveLoc[T.Medium]        1.9215      0.127     15.145      0.000       1.672       2.171
CompPrice                  0.0926      0.004     22.192      0.000       0.084       0.101
Income                     0.0159      0.002      8.569      0.000       0.012       0.020
Advertising                0.1148      0.008     14.694      0.000       0.099       0.130
Price                     -0.0953      0.003    -35.285      0.000      -0.101      -0.090
Age                       -0.0455      0.003    -14.145      0.000      -0.052      -0.039
==============================================================================
Omnibus:                        1.372   Durbin-Watson:                   1.994
Prob(Omnibus):                  0.504   Jarque-Bera (JB):                1.308
Skew:                           0.140   Prob(JB):                        0.520
Kurtosis:                       2.999   Cond. No.                     1.91e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.91e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

**Coefficients**: The "coef" column shows the actual model coefficients: 5.4684 for the intercept and 0.0159 for the Income variable, 0.1148 for Advertising variable, -0.0953 for Price variable, 4.8051 for ShelveLoc(Good) variable, 1.9215 for ShelveLoc (Medium) variable, 0.0926 for CompPrice variable and -0.0455 for Age variable.

Our model: **Sales**= 5.4684 + 0.0159*Income* + *0.1148*Advertising - 0.0953*Price* + *4.8051*ShelveLoc(Good) + 1.9215*ShelveLoc(Medium)* + *0.0926*CompPrice − 0.0455*Age+ e

**Significance of the variables**: At the 5% significance level, the intercept and all predictors (ShelveLoc Good, ShelveLoc Medium, CompPrice, Income, Advertising, Price and Age) are statistically significant, with p-values below 0.001. This indicates that each factor has a significant effect on sales.

**Interpretation** (Sales is measured in thousands of units; each effect holds other factors constant):

- **Shelf location** is the strongest driver. Compared with Bad placement, Good placement is associated with about 4,805 more units sold, and Medium placement with about 1,922 more units.
- **Price:** each $1 increase in price is associated with about 95 fewer units sold.
- **Competitor price:** each $1 increase in competitor price is associated with about 93 more units sold. This effect is almost equal and opposite to the store's own price effect, which suggests customers respond mainly to the price gap relative to competitors.
- **Advertising:** each additional $1,000 of local advertising is associated with about 115 more units sold.
- **Income:** each additional $1,000 of community income is associated with about 16 more units sold, a positive but relatively small effect.
- **Age:** each additional year in the local population's average age is associated with about 46 fewer units sold.

**Model fit**: The model explains 86.9% of the variation in sales across 400 stores (Adjusted $R^2$ = 0.866), using shelf location, competitor price, income, advertising, product price and age as predictors. The F-test ($p < 0.001$) confirms that the predictors are jointly significant.

The linear regression model explains 86.6% of sales variation across 400 stores using factors such as competitor price, income, advertising, population, product price, shelf display quality, location, age, and education.

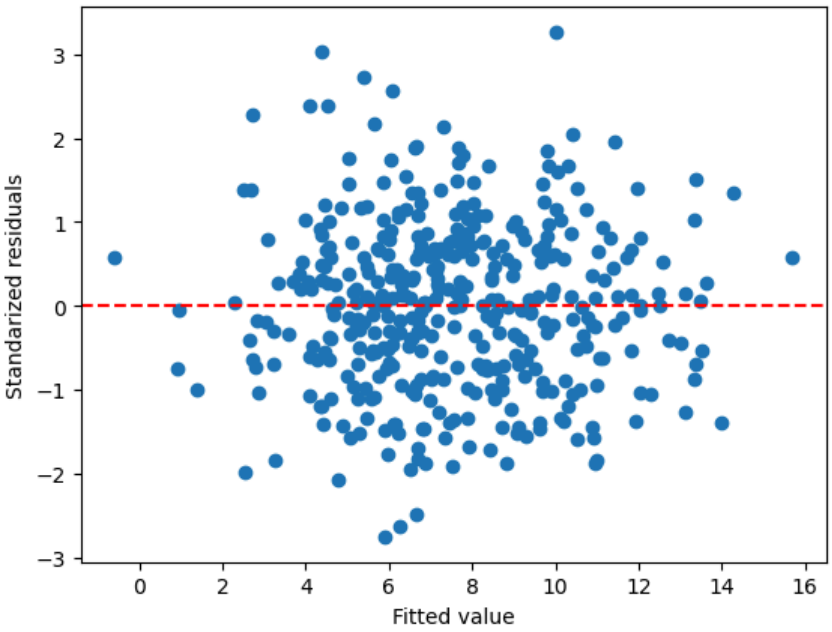
In [34]:
```python
# Check multicollinearity
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
X = model.model.exog
vif_df=pd.DataFrame({
    "Variable": model.model.exog_names,
    "VIF": [variance_inflation_factor (X,i) for i in range (X.shape[1])]
})
print(vif_df)
```

| | Variable | VIF |
|---|---|---|
| 0 | Intercept | 98.154154 |
| 1 | ShelveLoc[T.Good] | 1.497885 |
| 2 | ShelveLoc[T.Medium] | 1.498934 |
| 3 | CompPrice | 1.533554 |
| 4 | Income | 1.015391 |
| 5 | Advertising | 1.012818 |
| 6 | Price | 1.533286 |
| 7 | Age | 1.016798 |

All predictor variables have VIF values below 2 which is far below the common concern thresholds of 5 or 10. This shows that these variables are hardly strongly linearly related to each other and each variable contributes its own information to the model.

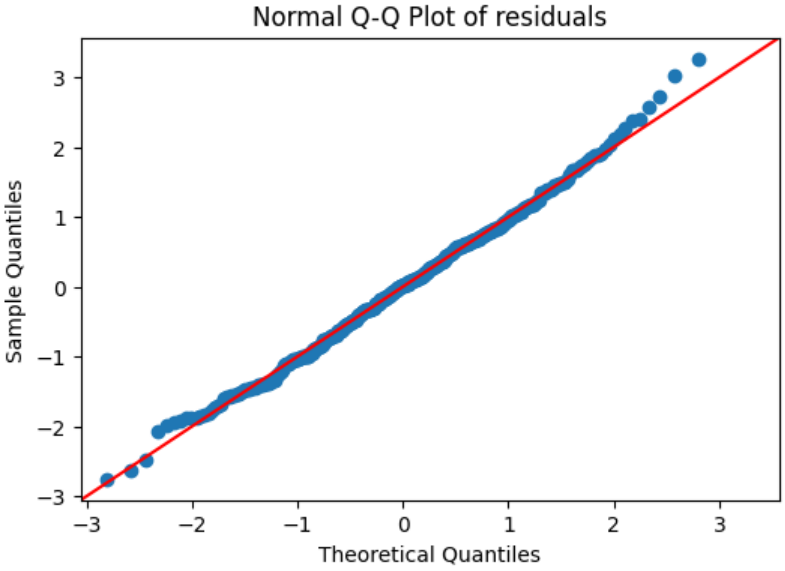
In [35]:
```
# Linear regression assumption
st_resids=model.get_influence().resid_studentized_internal
plt.scatter(model.fittedvalues,st_resids)
# Labeling each axis
plt.xlabel("Fitted value")
plt.ylabel("Standarized residuals")
plt.axhline(y=0, color='r', linestyle='--')
```

Out[35]: <matplotlib.lines.Line2D at 0x1f98857da90>




The residuals and fitted plot show residuals scattered randomly around zero with no visible pattern, indicating that the assumptions of linearity and constant variance are reasonably met.

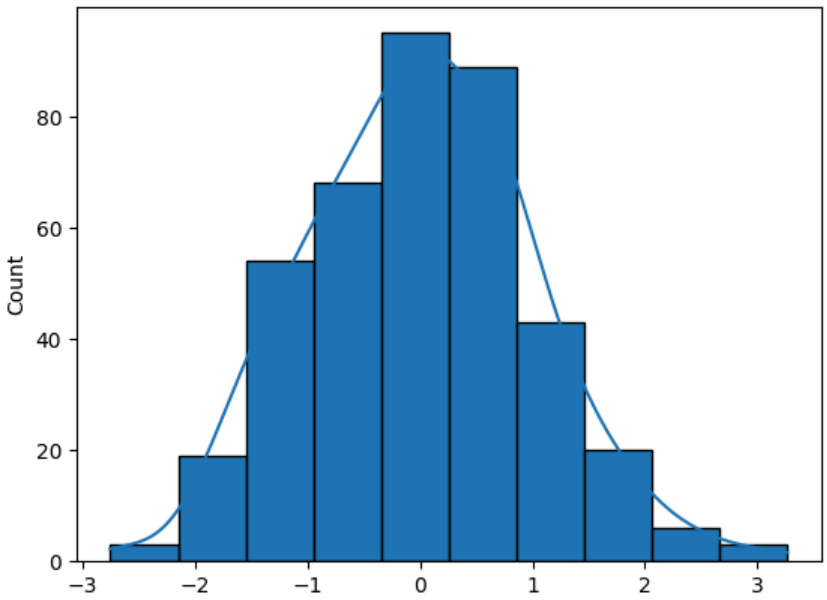
In [36]:
```
fig, ax = plt.subplots(figsize=(6, 4))
sm.qqplot(st_resids, line="45", ax=ax, fit=True)  # line="45" vẽ đường chéo chuẩn
ax.set_title("Normal Q-Q Plot of residuals")
ax.set_xlabel("Theoretical Quantiles")
ax.set_ylabel("Sample Quantiles")
plt.show()
```




The Q-Q plot shows most points lying close to the reference line, suggesting that the residuals are approximately normally distributed with only minor deviations at the tails.

In [37]:
```
# Histogram (Normality of error)
plt.hist(st_resids,bins=10)
sns.histplot(st_resids, bins = 10, kde=True)
```

Out[37]: <Axes: ylabel='Count'>




The histogram of standardized residuals also forms a roughly bell-shaped curve with most values concentrated around zero and symmetrically decreasing toward the tails, further supporting the normality assumption.

## 10. Conclusion

This project analysed the drivers of car seat sales across 400 stores using exploratory analysis, hypothesis testing and multiple linear regression. The final regression model explains 86.9% of the variation in sales (Adjusted $R^2$ = 0.866), and all diagnostic checks indicate that the key OLS assumptions are reasonably satisfied. The findings address the four questions set out in the introduction:

**1. Which store characteristics lead to higher performance?** Shelf location is the most influential factor. Holding other factors constant, stores with Good shelf placement sell about 4,805 more units than those with Bad placement, and Medium placement adds about 1,922 units. Local advertising also has a significant positive effect, with each additional 1,000 USD associated with about 115 more units sold.

**2. Do demographic features meaningfully impact sales?** Demographic effects are statistically significant but comparatively modest. Higher community income is linked to slightly higher sales (about 16 units per 1,000 USD), while an older local population is associated with lower sales (about 46 fewer units per additional year of average age). Urban location shows no meaningful difference in the sales distribution, and Population and Education show near-zero correlation with sales.

**3. Is there a difference in advertising level across regions?** Yes. The Chi-square test ($p < 0.001$) shows a strong association between advertising level and US location. Almost all US stores fall into the high advertising group, while most non-US stores spend little or nothing on local advertising.

**4. Are consumers sensitive to price?** Yes. Lower-priced stores achieve significantly higher sales in both high- and low-income areas (one-tailed t-tests, $p < 0.001$). The regression shows that each 1 USD increase in price reduces sales by about 95 units, while each 1 USD increase in competitor price raises sales by about 93 units. This suggests that customers respond mainly to the price gap relative to competitors.

**Recommendations**

- Prioritise securing Good shelf placement, as it delivers the largest sales gain of any factor.
- Set prices with close reference to local competitors rather than in isolation.
- Consider expanding advertising in non-US markets, where it is currently underused.

**Limitations** The data are cross-sectional, so the relationships identified are associations rather than proven causal effects. The model assumes linear relationships and does not test interaction effects, such as whether price sensitivity differs by shelf location. In addition, the median split used for income and price segments simplifies continuous variables and may hide more detailed patterns.